Faculty of Engineering & Information Technology
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31338 Network Servers

Lab 2d: Disk Partitioning and File System Management

Aims

1. Partition your hard disk

Task 1: Disk Management in CentOS

By default, Redhat and CentOS derivative Linux operating systems first create a disk partition called */boot*, just for the bootloader files. Then for the remaining disk space, they use the Logical Volume Manager to create 2 logical volume groups called *root* and *swap*. This allows the system to expand the size of the root drive as needed by simply adding more hard drives to the physical volume.

Open the Gnome app *Disks* from *System*.

What do you notice about the boot drive? Is it managed by the Logical Volume Manager? Why do you think it is done this way? (hint: Think about how you can boot before the LVM2 starts, or booting into single-user mode?)

On our lab image we have a simple 3 partition layout based on the default partition layout, /dev/nvme0n1, /dev/mapper/cs-root and /dev/mapper/cs-swap.

First thing would be to check the current layout. Use the following command

```
lsblk
```

to show how the current disk system is mapped.

Then run the command

```
df -h
```

to show the amount of free space in the mounted file system.

Task 2: Creating Partitions

We use the virtual partitions in VMWare. This requires shutting down the system.

Right-click the CentOS virtual machine in the left pane, then click *Settings*. In *Virtual Machine Settings*, you can set the properties of any listed virtual devices, e.g., enlarge the memory.

In the hardware lab, choose *Hard Disk*, and use the recommended *NVMe*. Then choose *Create a new virtual disk* and set up 2GB disk size. When this step is done, power on the virtual machine.

You can do the same thing on Windows Server.

We want to create 2 partitions:

- A 1GB EXT4 partition.

- A 1GB swap space partition

In the terminal, use the following command to see all partitions:

```
fdisk -l
```

Can you see a new partition /dev/nvme0n2?

The command fdisk is the default disk partition management tool with several options. Follow the instructions to see what the meanings of these options are by

```
fdisk -m
```

To start setting up a partition, use the command

```
fdisk <partition_id>
```

where partition_id is the full path of the partition, e.g., /dev/nvme0n2. Follow the prompts to finish the settings.

The new partitions need to be formatted, making them recognisable by the OS. Use the following two commands to format the disks.

```
mkfs -t <format> <partition_id>
mkswap <partition_id>
```

To turn on the swap, use *swapon* command

```
swapon <partition_id>
```

Now run the *top* command again. What does it say about swap space now?

Now let us mount the new ext4 partition on the /opt mount point.

Begin by creating some file in the /opt directory, e.g.,

```
touch /opt/hello.txt
```

Verify it using the *ls* command.

Then mount the new partition onto the /opt directory with the following command.

```
mount <partition_id> /opt
```

Now locate your hello.txt file again. Where is it?

Create a new file in /opt called "world.txt":

```
touch /opt/world.txt
```

Verify it with *ls*.

Unmount the partition with:

```
umount <partition_id>
```

Use *ls* command to look at what's in the /opt directory now. What's going on? Try mounting and unmounting a few times and check the /opt directory. Note your findings in your journal.

The above mounting is temporary. How to make it permanent, i.e., when the system reboots, the new partition created can be directly accessed from the mounting point /opt?

Task 3: Disk Management in Windows

Windows system manages disks using the internal software *Disk Management*. You can search it from the start menu.

Alternatively, Windows Server Manager has a separate dashboard for managing *File and Storage Services*. Find it in the left-hand menu of Server Manager. You should see four options: Servers, Volumes, Disks and Storage Pools. Start by choosing *Disks* and same window appears.

It shows that our system has a 60GB disk on the device *Disk 0*, with three partitions/volumes. The main volume is the one labelled *C:* which is the main OS hard drive. The new virtual device *Disk 1* has not been allocated.

Have a look around the way you did in Linux, and record notes in your learning journal about the partition structure.

Now right-click on the same drive again and choose *New Simple Volume*. In the pop-up wizard, follow the instructions to create a volume (partition) with drive letter *E* that uses all available space on the disk.

What are the formats available in Windows? Are they compatible with Linux systems?

Verify the new *E:/* drive exists, and its size.

Remember to take a snapshot of the virtual machines!